

Diffusion Models

Semester 2, 2024

Tom Drummond

Outline

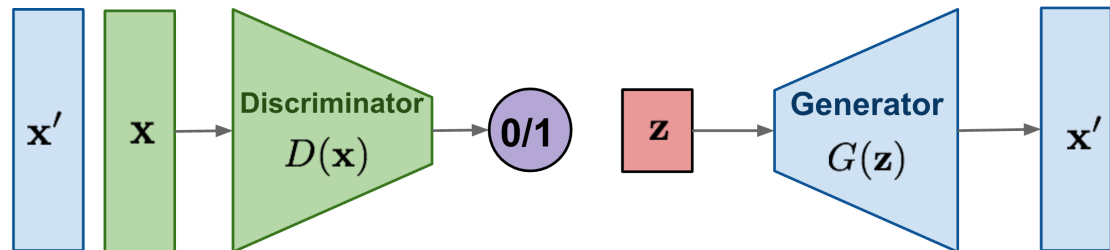
- Generative Models
- Image destruction as a diffusion process
- Inverting the process for image generation
- Diffusion for inpainting
- Diffusion on latent space (stable diffusion)
- Recent tricks to improve speed and quality

Generative Models

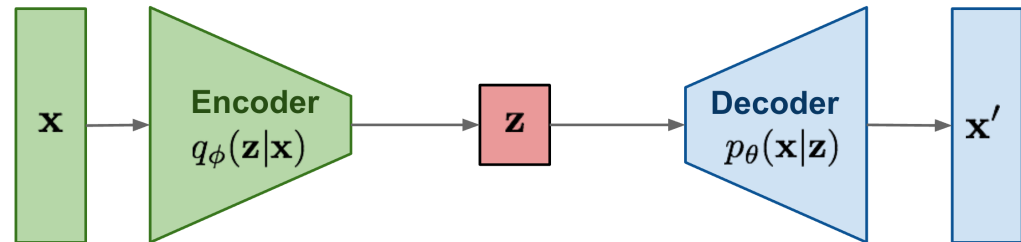
- Goal of (unconditional) generative models is to create a process for sampling from a distribution.
- For images this is hard because we have no way of writing down the probability that a set of pixel values is an image.
- But we do have samples of images!

Generative Models

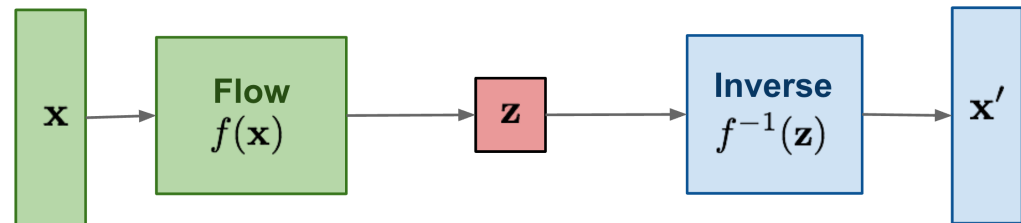
GAN: Adversarial training



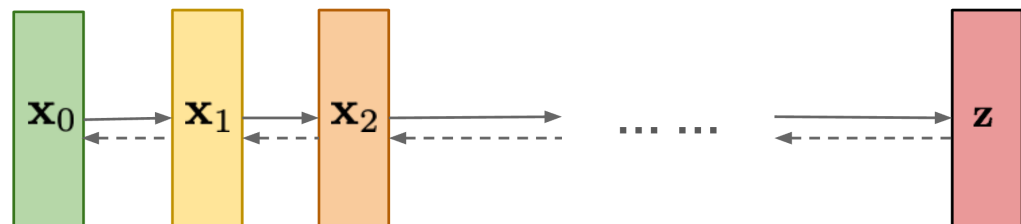
VAE: maximize variational lower bound



Flow-based models:
Invertible transform of distributions



Diffusion models:
Gradually add Gaussian noise and then reverse



Diffusion generated samples



Diffusion

- Diffusion models the destruction of a structured distribution as it moves towards Gaussian noise.
- Modelled as a random walk for each sample from the distribution. Update is a mixture between sample and random noise:

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon \quad \epsilon \sim N[0,1]$$

In 2D, each blue dot is a sample x_t

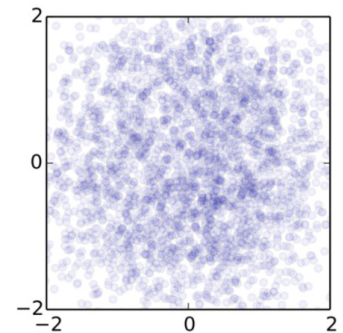
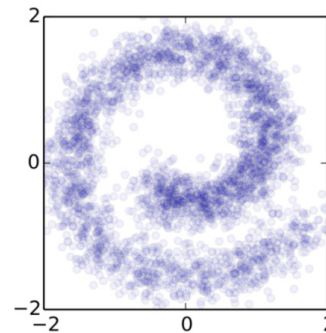
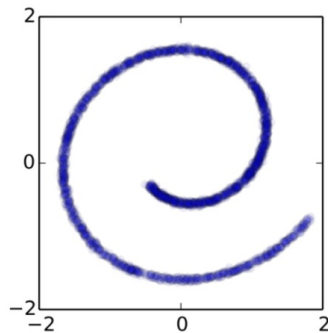


Image Diffusion

- Images live in a much higher dimension space
- But we can visualise the diffusion process applied to a single sample



x_0

x_1

x_2

x_3

x_4

x_5

$$x_t = \underbrace{\sqrt{1 - \beta_t} x_{t-1}}_{\mu} + \underbrace{\sqrt{\beta_t} \epsilon}_{\sigma} \quad \epsilon \sim N[0,1]$$

Multiple noising steps

- Multiple noising steps are just one large noising step

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon \quad \epsilon \sim N[0,1]$$

$$x_t = \sqrt{1 - \beta_t} \sqrt{1 - \beta_{t-1}} x_{t-2} + \sqrt{1 - \beta_t} \sqrt{\beta_{t-1}} \epsilon_{t-1} + \sqrt{\beta_t} \epsilon_t \quad \epsilon_t \sim N[0,1]$$

$$x_t = \sqrt{1 - \beta_t} \sqrt{1 - \beta_{t-1}} x_{t-2} + \sqrt{\beta_t + \beta_{t-1} - \beta_t \beta_{t-1}} \epsilon \quad \epsilon \sim N[0,1]$$

$$x_t = \sqrt{\bar{\alpha}_t} x_o + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad \bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$$

Image Diffusion

- Can rewrite the diffusion process as a distribution

$$x_t = \underbrace{\sqrt{1 - \beta_t} x_{t-1}}_{\mu} + \underbrace{\sqrt{\beta_t} \epsilon}_{\sigma} \quad \epsilon \sim N[0,1]$$

$$P(x_t | x_{t-1}) = N[\underbrace{\sqrt{1 - \beta_t} x_{t-1}}_{\mu}, \underbrace{\sqrt{\beta_t}}_{\sigma}]$$

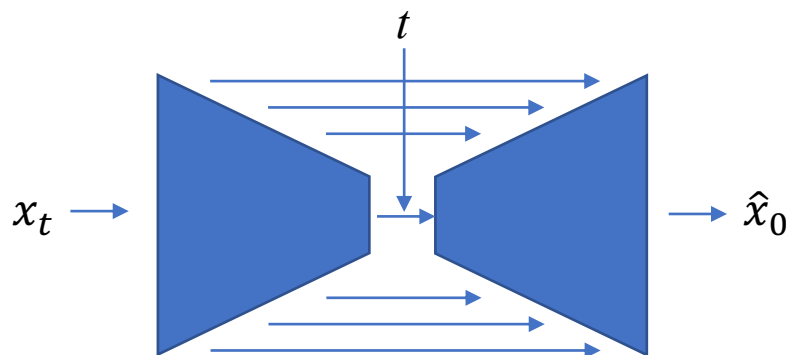
- Goal of generative process is to model reverse:

$$P(x_{t-1} | x_t)$$

but this is **not** Gaussian...

Using Deep Learning

Use a U-Net architecture to estimate x_0 given x_t and t

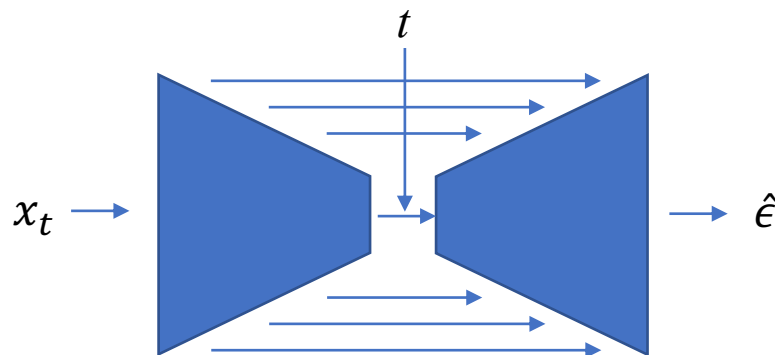


Can then estimate ϵ from x_0

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad \Rightarrow \quad \hat{\epsilon} = \frac{x_t - \sqrt{\bar{\alpha}_t} \hat{x}_0}{\sqrt{1 - \bar{\alpha}_t}}$$

Using Deep Learning

Or estimate ε given x_t and t



Can estimate x_0 from x_t and ε

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon \quad \Rightarrow \quad \hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \hat{\varepsilon}}{\sqrt{\bar{\alpha}_t}}$$

Using Deep Learning

Can now estimate x_{t-1}

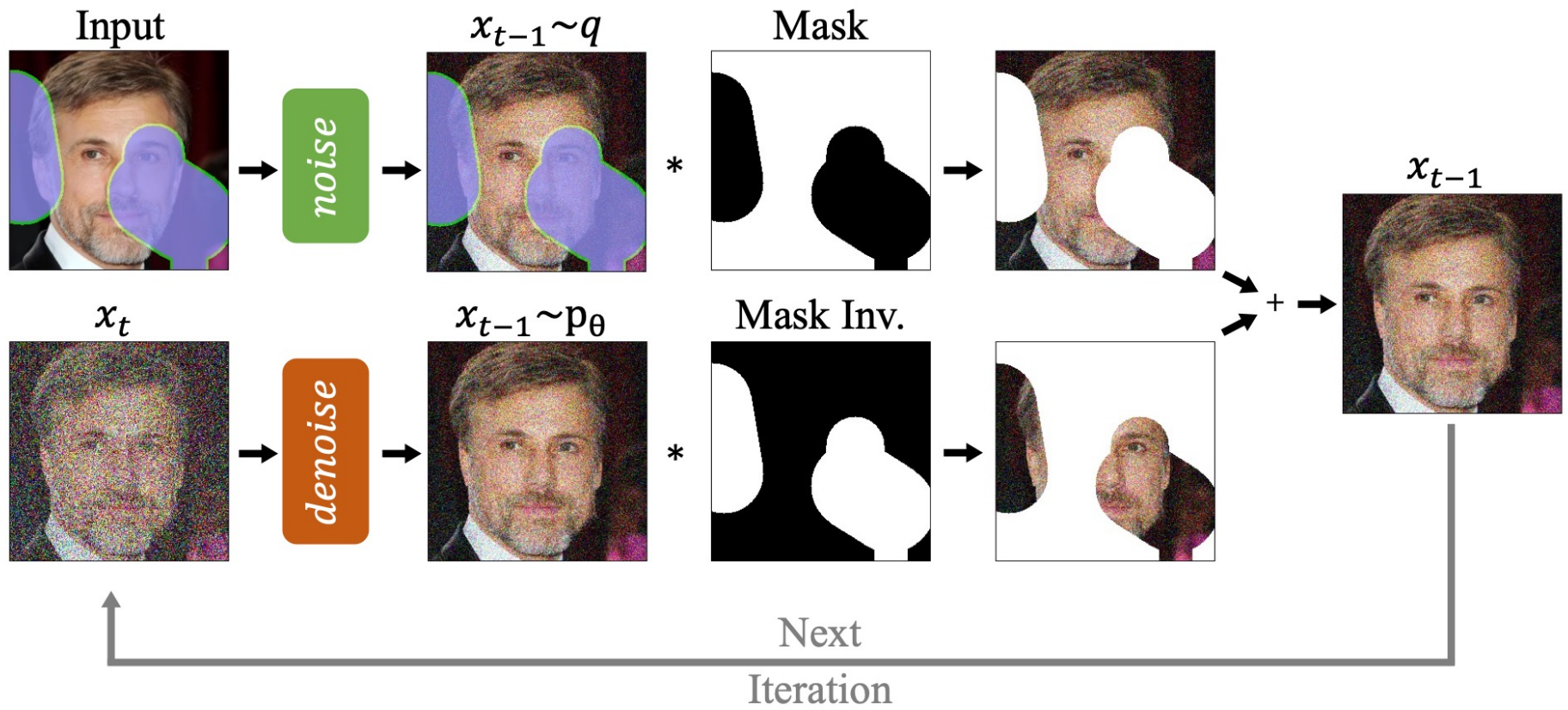
$$\hat{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\epsilon}$$

Can add noise to the estimate of x_{t-1}

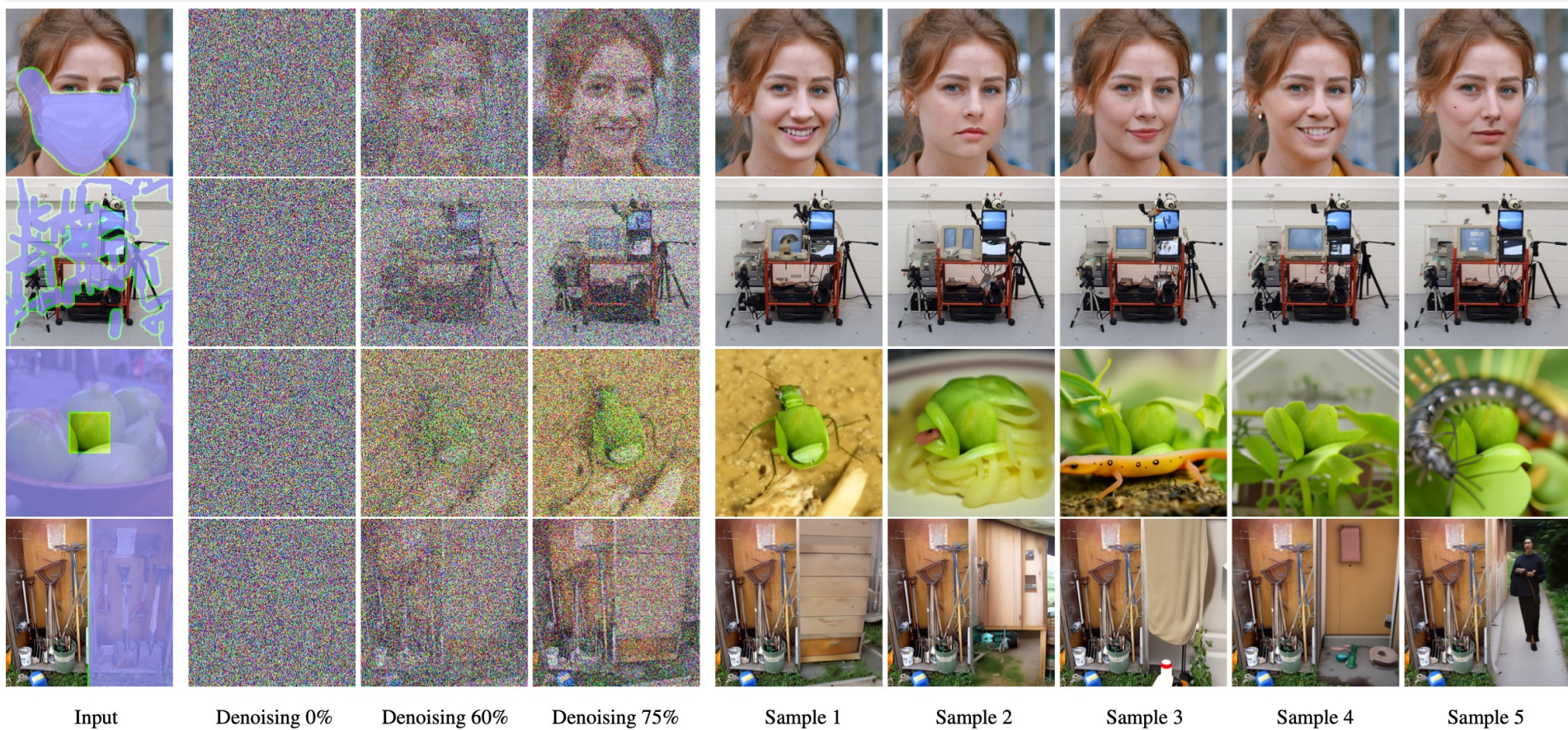
Or not! (this is called cold diffusion)

Good results typically take 1000 reconstruction steps

Diffusion for inpainting

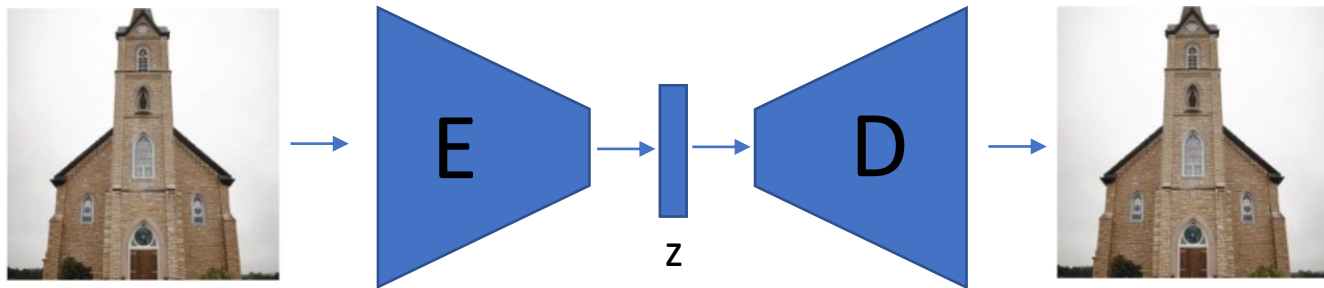


Diffusion for inpainting



Diffusion on latent space

Step 1: build Autoencoder



Step 2: Perform diffusion on latent space using z in place of x_0

Advantages:

- z is a very compact representation compared to the image
- Can use a smaller network to learn diffusion
- Decoder learns most regularities in the domain so diffusion model doesn't have to

Recent developments

Work by Zhenkai Zhang
(PhD student that Kris and I supervise)

$$x_t = \sqrt{\bar{\alpha}_t} x_o + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

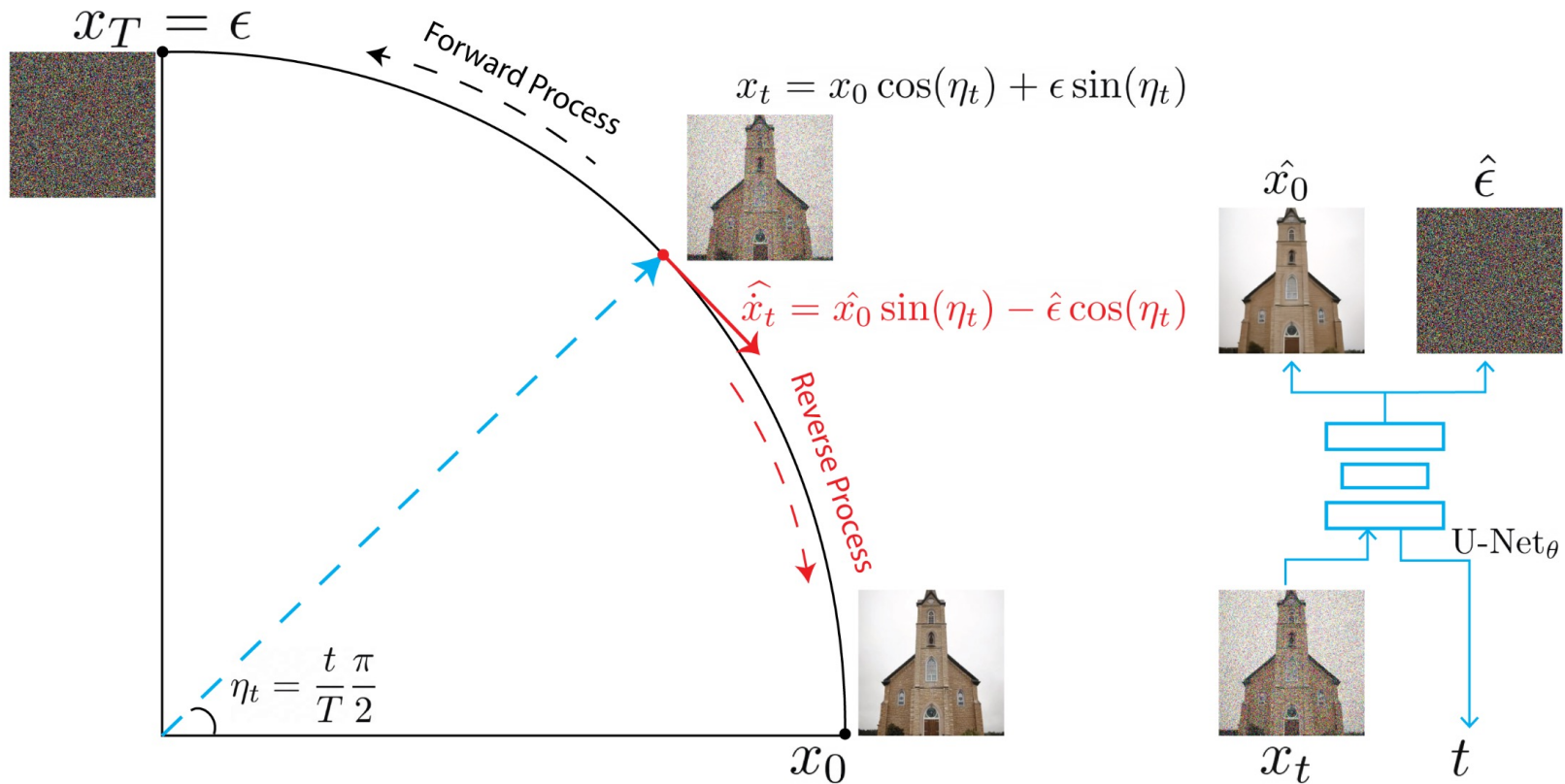
1) Reparameterise with cos and sin $x_t = \cos(\theta) x_o + \sin(\theta) \epsilon$

- This removes singularities at $\alpha = 0, 1$
- Cold reverse process can now be expressed as an Ordinary Differential Equation with $dx/dt = F(x, t)$
- Enables use of higher order ODE solvers, e.g. 4th order Runge-Kutta to find solutions in very few steps (e.g. 10).

2) Estimate both original image and noise: x_0 and ϵ

This is better than just estimating one quantity and computing the other.

Recent developments

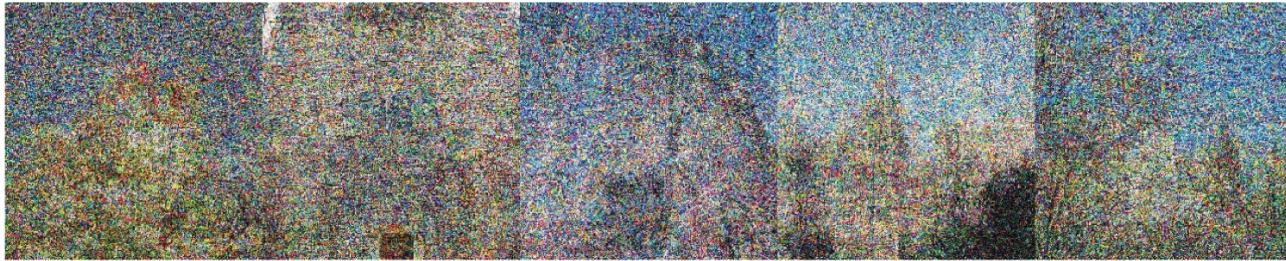


Also estimate both image and noise

Generation in 10 steps

steps = 10

DDPM



DDIM



Ours



Performance

FID scores

Steps	CIFAR-10					CelebA				
	10	16	20	50	100	10	16	20	50	100
DDIM	18.67	12.67	11.06	7.08	5.64	16.92	15.68	13.38	9.00	6.40
DDIM+ β^*	15.04	10.77	9.28	6.00	4.92	23.03	15.23	13.06	5.92	4.24
DDIM+sin()	15.97	11.90	9.79	6.22	5.10	24.71	17.18	11.87	5.25	4.22
DDIM+ $\hat{\mathbf{x}}_0, \hat{\epsilon}$	15.03	10.05	8.84	6.25	5.36	14.94	13.01	11.56	8.08	6.11
DDIM+sin() + $\hat{\mathbf{x}}_0, \hat{\epsilon}$	13.17	9.77	8.29	5.50	4.50	22.56	13.89	12.80	5.05	3.09
DDIM+ β^* + $\hat{\mathbf{x}}_0, \hat{\epsilon}$	13.69	10.19	8.73	5.80	4.73	23.92	12.58	12.67	4.72	2.83
Ours	9.50	6.32	5.62	4.57	4.17	22.70	8.80	11.00	4.05	2.39